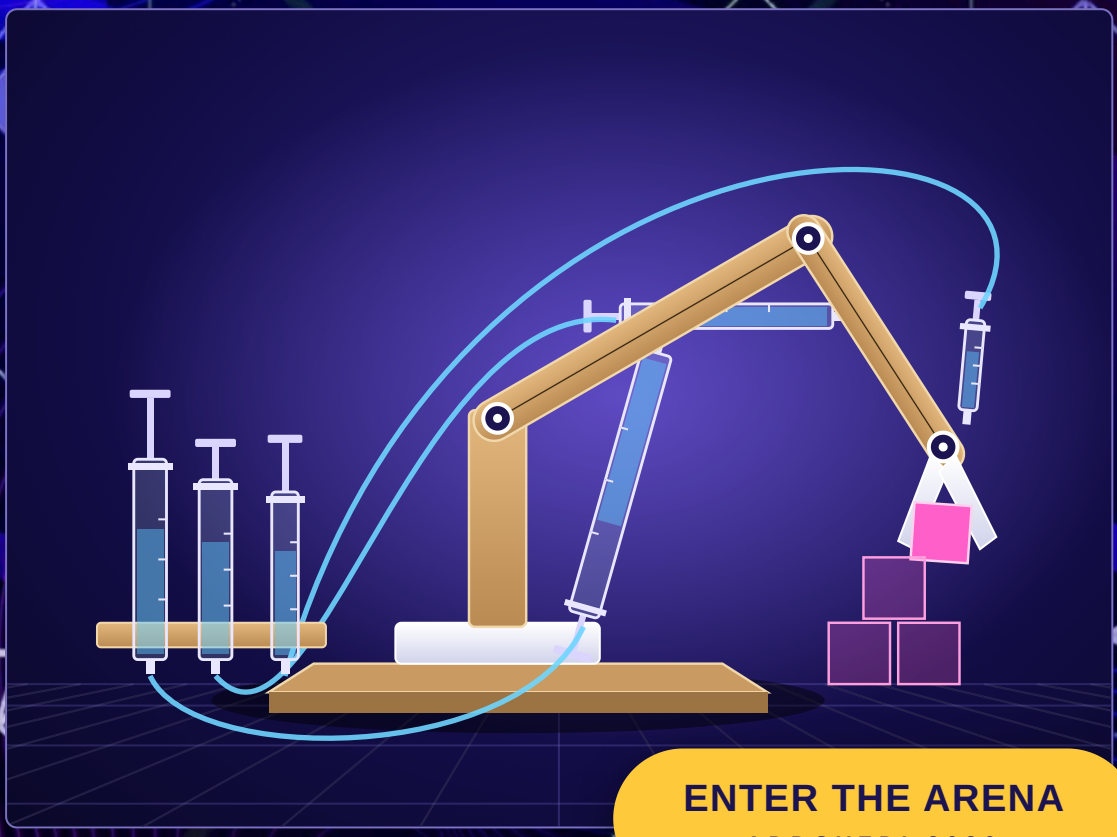
NIT ARUNACHAL PRADESH

ADDOVEDI

ENTER THE ARENA

HYDRAULIC WARZONE

HYDRAULIC ARM CHALLENGE • RULEBOOK



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ADDOVEDI 2026

ORGANISED BY

Mechanical Engineering Student's Society (MESS)

National Institute of Technology Arunachal Pradesh

ADDOVEDI 2026 • Theme: Enter the Arena

1. Event Overview

Hydraulic Warzone is the hydraulic arm challenge of ADDOVEDI 2026, organised by the Mechanical Engineering Student's Society (MESS), National Institute of Technology Arunachal Pradesh. It is a hands-on mechanical engineering competition in which participants design, build and operate a manually controlled hydraulic robotic arm.

Teams construct their hydraulic arm from the permitted materials and then operate it to complete a series of practical challenges inside the competition arena.

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WHAT THE COMPETITION TESTS

AREA	WHAT PARTICIPANTS DEMONSTRATE
Hydraulic systems	Understanding of how pressure is transmitted through a fluid-filled syringe system
Mechanical linkages	Design of joints and links that convert syringe motion into arm movement
Structural design	A rigid, well-proportioned base and arm that can carry loads
Hydraulic actuation	Effective use of syringes as actuators for each degree of movement
Precision and control	Smooth, deliberate movement and accurate placement of objects
Stability	An arm and base that remain steady during lifting and gripping
Problem-solving	Quick diagnosis and correction of mechanical or hydraulic issues
Teamwork	Coordinated building and operation by all three members

2. Team Formation

3
MEMBERS PER TEAM

₹100
REGISTRATION FEE / TEAM

Open
NO LIMIT ON TEAMS

2.1 Each team must consist of **exactly 3 members**.

2.2 There is **no maximum limit** on the total number of teams that can register for the event.

2.3 The registration fee is **₹100 per team**.

2.4 Each participant may be a member of only one team.

2.5 All team members must complete the registration process.

2.6 Participants must provide valid and accurate registration details. Incorrect or misleading details may lead to cancellation of the registration.

2.7 Team composition cannot be changed after registration without permission from the organisers.

3. Component Allocation

The organisers will provide hydraulic arm construction kits to a limited number of teams. Kits are allotted only to the **first 8 successfully registered teams**. Registration for the event itself remains open to all teams.

3.1 Component kits will be distributed strictly on a **first come, first served** basis.

3.2 A team becomes eligible for a component kit only after completing both registration and payment.

3.3 Only **8 component kits** are available.

3.4 Once all 8 kits have been allocated, no further organiser-provided kits will be available.

3.5 Teams registering after the first 8 may still register for and take part in the event, but they will not receive the organiser-provided component kit.

3.6 The organisers will announce when all 8 kits have been allotted.

3.7 The provided kit is intended solely for constructing the hydraulic arm for the competition.

TEAM STATUS	KIT PROVISION
Registered and paid within the first 8	Receives the organiser-provided component kit
Registered after all 8 kits are allotted	Participates in the event without an organiser-provided kit

4. Components Provided by Organisers

The organiser-provided hydraulic arm construction kit contains the following components. No additional components are part of the kit.

CATEGORY	COMPONENT	PURPOSE IN THE HYDRAULIC ARM
Structural materials	Plywood	Rigid structural parts such as the base, support column and load-bearing arm links
Structural materials	Large Thermocol	Lightweight structural parts, spacers, linkages and gripper elements
Hydraulic components	10 mL small syringes	Hydraulic actuators and control syringes for joints that need finer, controlled movement
Hydraulic components	20 mL syringes	Hydraulic actuators and control syringes for joints that need longer stroke or more displacement

CATEGORY	COMPONENT	PURPOSE IN THE HYDRAULIC ARM
Hydraulic components	Hydraulic tube / tubing	Connects paired syringes and transfers pressure from the operator's syringe to the actuator syringe
Joining materials	Fevicol	Bonding structural parts, especially plywood and Thermocol joints
Joining materials	Super Glue	Quick, strong bonding of small parts and fixing syringes in position
Construction tool	Paper Cutter	Cutting and shaping the permitted construction materials

HOW THE HYDRAULIC SYSTEM WORKS

Each movement of the arm is produced by a pair of syringes connected by tubing and filled with fluid. When the operator pushes or pulls the plunger of the control syringe, the fluid transfers that motion through the tube to the actuator syringe mounted on the arm, which then moves the linked joint.

5. Component Usage Rules

5.1 Teams must construct the hydraulic arm primarily using the materials provided by the organisers.

5.2 All movement of the arm must be generated through the syringe-based hydraulic mechanism.

5.3 Syringes must be securely mounted to the structure.

5.4 Tubes must be properly connected to the syringes.

5.5 The hydraulic system must not have significant leakage.

5.6 Participants may creatively design the geometry, structure, linkages, base and gripping mechanism using the permitted materials.

5.7 Participants must not intentionally damage or misuse the provided components.

5.8 The hydraulic arm must be **manually operated**.

PROHIBITED ITEMS AND SYSTEMS

Unless explicitly permitted by the organisers, the following are prohibited:

- Electric motors
- Batteries
- Automated control systems
- Pneumatic systems
- Any motorised mechanism
- Servo motors
- Electronic controllers
- External powered actuators
- High-pressure hydraulic systems
- Any mechanism that provides an unfair advantage

6. Build and Preparation

Teams will receive the permitted materials and construct their hydraulic arm. The construction stage may involve the following steps:

#	STEP	FOCUS
1	Designing the base	A stable footprint that resists tipping during operation
2	Designing the arm structure	Link lengths and proportions suited to the tasks
3	Creating mechanical linkages	Joints that pivot freely without wobble
4	Installing hydraulic syringes	Firm mounting at the correct positions on each joint
5	Connecting hydraulic tubes	Secure, leak-free connections between syringe pairs
6	Creating a gripping mechanism	Reliable opening, closing and holding of objects
7	Testing joint movement	Full, smooth range of motion at every joint
8	Testing lifting capability	Raising objects without slipping or sagging
9	Testing gripping capability	Holding objects securely during movement
10	Ensuring structural stability	No loose parts, cracks or weak joints

Participants are encouraged to use engineering creativity while staying within the permitted materials and rules.

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7. Qualification / Testing Round

Before the main challenge, every team's hydraulic arm will undergo a testing round conducted by the organisers.

CHECK	WHAT IS VERIFIED
Basic hydraulic movement	The arm responds to the control syringes
Lifting capability	The arm can raise an object
Gripping capability	The gripper can grasp and hold an object
Joint movement	Each joint moves through its intended range
Structural stability	The base and arm stay steady during operation
Hydraulic functionality	Syringes and tubes work without significant leakage

7.1 The organisers may reject, or temporarily stop, an unsafe or non-functional arm until the issue is corrected.

7.2 Only teams that satisfy the basic functionality and safety requirements will proceed to the main challenge.

8. Main Challenge

Qualified teams will enter the Hydraulic Warzone arena. The hydraulic arm may be required to perform practical tasks such as the following:

A **PICK AND PLACE**
Grip, lift and position
Pick up a designated object and place it at a specified location.

B **OBJECT TRANSFER**
Controlled movement
Move objects from one location to another.

C **STACKING**
Vertical placement
Pick up objects and stack them in a specified arrangement.

D **OBSTACLE / RESTRICTED ZONE CHALLENGE**
Spatial control
Operate the arm while dealing with designated obstacles or restricted areas.

E **PRECISION TASK**
Accuracy
Perform a controlled placement or manipulation task requiring accuracy.

The exact arena layout, objects, task sequence and task-specific scoring will be announced by the organisers before the competition.

9. Competition Procedure

The competition follows the general flow below.

STAGE 1 **Registration** → STAGE 2 **Component Allocation** → STAGE 3 **Build & Preparation** → STAGE 4 **Testing / Qualification**

STAGE 5 **Main Arena Challenge** → STAGE 6 **Judging** → STAGE 7 **Final Score** → STAGE 8 **Winner Declaration**

STAGE	DESCRIPTION
1. Registration	Teams of exactly 3 members register and pay the ₹100 registration fee. Registration is open to any number of teams.
2. Component Allocation	The first 8 teams to complete registration and payment receive the organiser-provided component kit, on a first come, first served basis.
3. Build & Preparation	Teams design and construct their hydraulic arm using the permitted materials, then test and refine it.
4. Testing / Qualification	The organisers check each arm for basic movement, lifting, gripping, stability and hydraulic functionality. Arms meeting the functionality and safety requirements qualify.
5. Main Arena Challenge	Qualified teams operate their arm in the arena to complete the announced tasks within the time limit.

STAGE	DESCRIPTION
6. Judging	Judges evaluate each attempt against the scoring criteria and apply any penalties.
7. Final Score	Marks for all criteria are totalled (out of 100) after penalties; tie-breaking rules are applied where required.
8. Winner Declaration	The organisers announce the final rankings and the prize winners.

10. Time Limit

The exact time limit for the main challenge will be announced by the organisers before the competition.

10.1 The timer starts on the official signal given by the judges.

10.2 Teams must stop operating the arm immediately when time expires.

10.3 Tasks completed after the official time limit will not be counted.

10.4 Additional time will be granted only if the organisers cause an interruption.

10.5 Judges may stop an attempt at any time for safety reasons.

11. Scoring and Judging

Each team's performance in the main challenge will be evaluated out of **100 marks** as follows.

CRITERION	WHAT IS EVALUATED	MARKS
Successful Task Completion	Marks for successfully completing the assigned tasks	**40**
Accuracy & Precision	Accurate placement, controlled gripping and correct positioning of objects	**25**
Hydraulic Arm Control	Smooth and effective operation of the hydraulic mechanism	**15**
Time Efficiency	How efficiently the team completes the tasks within the allotted time	**10**
Stability & Structural Performance	Stability of the base, arm, joints, hydraulic connections and overall construction	**10**
Total		**100**

CRITERIA EXPLAINED

11.1 **Successful Task Completion (40):** awarded for each assigned task that is completed successfully as defined by the judges.

11.2 **Accuracy & Precision (25):** evaluates how accurately objects are placed, how well they are gripped without dropping, and how precisely the arm is positioned.

11.3 **Hydraulic Arm Control (15):** evaluates the smoothness, responsiveness and effectiveness of the hydraulic operation, including coordination between the joints.

11.4 **Time Efficiency (10):** evaluates how efficiently the team completes the tasks within the allotted time.

11.5 **Stability & Structural Performance (10):** evaluates the stability of the base, arm, joints and hydraulic connections, and the overall quality of construction during the attempt.

The organisers may adjust task-specific scoring depending on the final arena configuration. The final scoring system will be communicated to all teams before the competition begins.

12. Penalties

Violations during the competition will attract the consequences listed below. The severity of each penalty will be decided by the judges based on the nature of the violation.

#	VIOLATION	POSSIBLE CONSEQUENCE
1	Directly touching competition objects when the hydraulic arm is supposed to manipulate them	Task cancellation or score deduction
2	Manually moving the hydraulic arm instead of using hydraulic actuation	Task cancellation or score deduction
3	Moving or modifying the arena	Score deduction or attempt stoppage
4	Interfering with another team	Score deduction or disqualification
5	Using unauthorised components	Disqualification
6	Using unauthorised external assistance	Score deduction or disqualification
7	Continuing after the time limit	Tasks not counted; score deduction
8	Unsafe operation	Attempt stoppage or disqualification
9	Deliberate damage to competition equipment	Disqualification
10	Misconduct	Score deduction or disqualification

TYPES OF CONSEQUENCE

CONSEQUENCE	MEANING
Task cancellation	The affected task is not counted
Score deduction	Marks are deducted from the team's score
Attempt stoppage	The team's attempt is stopped by the judges
Disqualification	The team is removed from the competition

13. Safety Rules

13.1 Participants must follow all instructions given by the organisers and safety officials.

13.2 Operate the hydraulic arm carefully and in a controlled manner at all times.

13.3 Keep hands and fingers away from moving joints and gripping mechanisms during operation.

13.4 Ensure that all hydraulic tubes and syringes are securely connected before operating the arm.

13.5 Report any leakage or damaged component to the organisers immediately.

13.6 Do not apply excessive force to the hydraulic system, syringes or tubes.

13.7 Sharp or dangerous modifications to the arm are prohibited.

13.8 Handle the paper cutter with care, use it only for cutting permitted materials, and keep it closed when not in use.

13.9 Do not operate the arm in any way that could injure participants, officials or spectators.

13.10 Organisers may stop an unsafe attempt immediately.

13.11 Decisions of the safety officials are final.

14. Tie-Breaking

If two or more teams have the same final score, the tie will be resolved in the order below. The team ranking higher at the first criterion where they differ is placed higher.

COMPARE FIRST **Tasks completed** → COMPARE NEXT **Accuracy score** → COMPARE NEXT **Time efficiency** → IF STILL TIED **Tie-break task**

14.1 The team with the greater number of successfully completed tasks ranks higher.

14.2 If still tied, the team with the higher Accuracy & Precision score ranks higher.

14.3 If still tied, the team with the better Time Efficiency score ranks higher.

14.4 If still tied, the organisers may conduct an additional tie-breaking task.

15. Disqualification

A team may be disqualified from the competition for any of the following:

15.1 Using prohibited components.

15.2 Using motorised, electronic or automated assistance.

15.3 Deliberately damaging the arena or equipment.

15.4 Interfering with another team.

15.5 Tampering with another team's equipment.

15.6 Cheating of any kind.

15.7 Repeated safety violations.

15.8 Misconduct.

15.9 Any action that compromises the fairness or safety of the competition.

16. General Rules

16.1 Participants must report according to the schedule announced by the organisers.

16.2 All teams must attend the mandatory briefing if instructed.

16.3 Teams are responsible for the safe handling of their components.

16.4 Maintain cleanliness in the competition area.

16.5 Do not enter restricted areas.

16.6 Do not interfere with another team's build or testing.

16.7 Maintain discipline and sportsmanship at all times.

16.8 Organisers may inspect any hydraulic arm at any stage of the competition.

16.9 Any unsafe or unfair mechanism may be rejected.

16.10 Organisers may modify the arena layout or task configuration if required.

16.11 Any such changes will be announced before the relevant round.

16.12 Judges' decisions regarding task validity, penalties and final rankings are final and binding.

17. Prize Pool

Prize Value: ₹13,000+ (including goodies)

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18. Final Authority

18.1 The organisers reserve the right to interpret the rules in this rulebook.

18.2 Any situation not explicitly covered in this rulebook will be decided by the organising committee and judging panel.

18.3 Decisions regarding safety, task validity, penalties and final rankings will be final.

18.4 Any rule modification will be communicated to participants before it is implemented.

Contacts

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SCAN TO JOIN THE WHATSAPP GROUP

Best of luck to all participants!

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